MODULE DESCRIPTION FORM

نموذج وصف المادة الدراسية

Module Information معلومات المادة الدراسية					
Module Title	**Advanced Programming Language**			**Module Delivery**	
Module Type	**Core**			☒ **Theory** ☒ **Lecture** ☒ **Lab** ☒ **Tutorial** ☒ **Practical** ☐ **Seminar**	
Module Code	**COMP102**				
ECTS Credits	**7**				
SWL (hr/sem)	**175**				
Module Level		1	**Semester of Delivery**		2
Administering Department		CS	**College**	CS and Mathematics	
Module Leader	Name		**e-mail**		
Module Leader's Acad. Title		Assistant Professor	**Module Leader's Qualification**		Ph.D.
Module Tutor			**e-mail**		
Peer Reviewer Name		Name	**e-mail**	E-mail	
Scientific Committee Approval Date		01/06/2023	**Version Number**	1.0	

Relation with other Modules العلاقة مع المواد الدراسية الأخرى			
Prerequisite module	Programming Fundamentals	**Semester**	1
Co-requisites module	Object-Oriented Programming	**Semester**	3
Co-requisites module	Data Structure	**Semester**	3
Co-requisites module	Website Design	**Semester**	5

Module Aims, Learning Outcomes and Indicative Contents أهداف المادة الدراسية ونتائج التعلم والمحتويات الإرشادية	
Module Objectives أهداف المادة الدراسية	1. This module aims 2. Optimal use of one-dimensional arrays to store data and control indexes. 3. Optimal use of two-dimensional arrays to store data and control indexes. 4. Teaching students how to use Java method and call it whatever it is needed in the correct position in the program 5. To study different types of Java methods 6. Using Classes to provide reusability, and managing objects to support inheritance modularity and information hiding.
Module Learning Outcomes مخرجات التعلم للمادة الدراسية	7. Knowledge and understanding 8. Understanding the syntax for declaring and allocating one- and two-dimensional arrays, in addition to creating user-defined methods and knowing how to call them. 9. Cognitive skills (thinking and analysis). 10. After understanding the syntax of programming tools, the student will have the main requirements for analysis problems and solve them using Java code. 11. Communication skills (personal and academic). 12. This course teaches the student to work within a group. 13. Practical and subject-specific skills (Transferable Skills). 14. Learn the main function in Java language for advanced programming tools like arrays , methods, and classes.
Indicative Contents المحتويات الإرشادية	This course will provide new tools for Java programming. How to deal with arrays is the main goal of this course. Using methods in different ways and for different uses is also one of the goals of the course. Learning how to collect data of one type with one array and how to reduce code size using methods is another goal. Attempting to merge the programming tools of the previous course and the current tools to remind them constantly is the last aim of the course.

Learning and Teaching Strategies استراتيجيات التعلم والتعليم	
Strategies	The main strategy that will be adopted in delivering this module is to encourage students' participation in the exercises, while at the same time refining and expanding their critical thinking skills. This will be achieved through classes, interactive tutorials, practicals, lab hours and by considering types of some small projects that are interesting to the students.

Student Workload (SWL) الحمل الدراسي للطالب محسوب لـ ١٥ اسبوعا			
Structured SWL (h/sem) الحمل الدراسي المنتظم للطالب خلال الفصل	110	**Structured SWL (h/w)** الحمل الدراسي المنتظم للطالب أسبوعيا	7
Unstructured SWL (h/sem) الحمل الدراسي غير المنتظم للطالب خلال الفصل	65	**Unstructured SWL (h/w)** الحمل الدراسي غير المنتظم للطالب أسبوعيا	6
Total SWL (h/sem) الحمل الدراسي الكلي للطالب خلال الفصل	**175**		

Module Evaluation تقييم المادة الدراسية					
		Time/Number	**Weight (Marks)**	**Week Due**	**Relevant Learning Outcome**
Formative assessment	**Quizzes**	2	10% (10)	5 and 10	All
	Assignments	2	10% (10)	4 and 12	All
	Projects / Lab.	1	10% (10)	Continuous	All
	Report	1	10% (10)	13	All
Summative assessment	**Midterm Exam**	2hr	10% (10)	7	All
	Final Exam	3hr	50% (50)	16	All
Total assessment			100% (100 Marks)		

Delivery Plan (Weekly Syllabus) المنهاج الاسبوعي النظري	
	Material Covered
Week 1+2	Java String: Types of java methods on string with examples
Week 3+4	One Dimensional Array with examples on one dimensional arrays
Week 5+6	Two dimensional Array with examples on two dimensional arrays
Week 7	First Exam
Week 8	Java Methods: The importance of using methods and methods-based programming structure. Methods return types, access specifiers and parameters
Week 9	Writting methods and calling them
Week 10+11	Sending and recieving datatypes between methods (Return and receive referenced and value data)
Week 12	Examples on Java Methods
Week 13	Second Exam

Week 14	Revision and project discussion
Week 15	Final Exam

Delivery Plan (Weekly Lab. Syllabus) المنهاج الاسبوعي للمختبر	
	Material Covered
Week 1	Lab 1: Write program to apply all methods of string
Week 2	Lab 2: Simple java program on static method
Week 3	Lab 3: writes programs on 1 D array
Week 4	Lab 4: solve assignments programs on 1 D array
Week 5	Lab 5: writes programs on 2 D array
Week 6	Lab 6: solve assignments programs on 2 D array
Week 7	Lab 7: write all examples on 1D and 2D arrays
Week 8	First Exam
Week 9	Lab 8: write programs on method with no return value and no pass parameter
Week 10	Lab 9: write programs on method with no return value and pass parameter
Week 11	Lab 10: write programs on method with return value and pass parameter
Week 12	Lab 11: Second Exam
Week 13	Lab 12: Write java programs in assignments
Week 14	Lab 12: Discussion assignments
Week15	Final Exam

Learning and Teaching Resources مصادر التعلم والتدريس		
	Text	**Available in the Library?**
Required Texts	**Java How to Program** (9th Edition) ,Paul and Harvey Deitel,2011.	Yes
Recommended Texts	**Java for Absolute Beginners**, Iuliana Cosmina, 2018	Yes
Websites	https://www.w3schools.com/java/ https://www.codecademy.com/learn/learn-java https://www.javatpoint.com/java-tutorial	

Grading Scheme مخطط الدرجات				
Group	**Grade**	التقدير	**Marks %**	**Definition**
Success Group (50 - 100)	**A -** Excellent	امتياز	90 - 100	Outstanding Performance
	B - Very Good	جيد جدا	80 - 89	Above average with some errors
	C - Good	جيد	70 - 79	Sound work with notable errors
	D - Satisfactory	متوسط	60 - 69	Fair but with major shortcomings
	E - Sufficient	مقبول	50 - 59	Work meets minimum criteria
Fail Group (0 – 49)	**FX –** Fail	راسب (قيد المعالجة)	(45-49)	More work required but credit awarded
	F – Fail	راسب	(0-44)	Considerable amount of work required

Note: Marks Decimal places above or below 0.5 will be rounded to the higher or lower full mark (for example a mark of 54.5 will be rounded to 55, whereas a mark of 54.4 will be rounded to 54. The University has a policy NOT to condone "near-pass fails" so the only adjustment to marks awarded by the original marker(s) will be the automatic rounding outlined above.